

Faculty of New Technologies of Information and Communication

Specifications for Android Application: Mindful Parenting App

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A Specifications Report for Android Application: Mindful Parenting App Submitted Using Scrum as Agile Methodology to the Software Technologies and Information Systems Program of University of Abdelhamid Mehri - Constantine 2 in Partial Fulfillment of the Requirements for the Master's of Software Engineering and Smart Systems Course of

Agile Development

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May 2024

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Introduction

Motivation

The Agile methodology revolutionizes project management by dividing projects into phases and promoting continuous collaboration and improvement. Unlike the traditional ”waterfall” approach, Agile emphasizes the importance of cross-functional teams working together seamlessly. Open communication, collaboration, adaptation, and trust among team members are central tenets of Agile, fostering a dynamic and responsive work environment.

In Agile, teams follow a cycle of planning, executing, and evaluating, allowing for quick adjustments based on feedback. While the project lead or product owner sets priorities, the team takes ownership of how tasks are executed, self-organizing around granular assignments. This approach enables teams to respond rapidly to changes in the marketplace or customer feedback without derailing long-term plans.

SCRUM

Scrum is an agile project management framework designed to assist teams in structuring and managing their work using a set of values, principles, and practices. Much like a rugby team preparing for a match, scrum encourages teams to learn from their experiences, self-organize while tackling challenges, and reflect on their successes and failures for ongoing enhancement.

While scrum aligns with agile’s principle of continuous improvement, some confuse the two since scrum embodies this core principle. Yet, the latter serves as a framework for accomplishing tasks, while agile represents a broader philosophy. Agile philosophy revolves around continual, incremental improvement achieved through frequent and small releases.

In this specifications report, we will explore the aspects of SCRUM through a case study and highlight the main aspects of it.

Chapter 1

Case Study

To delve deeper into the workings of the Agile method 'Scrum,' along with its nuances and benefits, we will exemplify it through our very own project—a mindful parenting Android app. This project, undertaken by a team of four members, serves as a real-life case study, showcasing the application of Scrum methodology. Through this illustration, we aim to explore the different stages of Scrum and the collaborative tools employed in our project.

1.1 Context and Project Definition

Android mobile application to address the modern challenges faced by parents in fostering emotional intelligence and connection with their children amidst the digital age through features like guided meditation sessions, daily reflection prompts, and parenting tips.

1.2 Project Objectives

- Cultivating a positive parenting mindset.
- Prioritizing intentional time for family activities.
- Raising awareness about correct parenting practices in the community.
- Establishing partnerships with doctors and concerned organizations.
- Creating a more supportive environment for children.

1.3 Project Scope

- Parents and parental figures with children under 13.
- Experts and specialized organizations.
- Individuals seeking to teach themselves mindfulness techniques to eventually integrate them into their parenting techniques.

1.4 Functional Description of Requirements

The app is divided into four different parts/sections as follows:

1.4.1 Feedback and account creation section

- All users can create accounts and manage their profiles.
- All users can access the feedback system.

1.4.2 Parenting section

- Parents can customize exercises (like meditation, stress reduction, etc...) according to their preferences.
- Parents can track their progress with the exercises they engage in.
- Parents can make daily notes and do their journaling in the app.
- Parents can set personalized reminders.
- Parents can search for and find experts' profiles.
- Parents can join discussions.
- Parents can read articles written by experts and engage with them (likes, comments).
- Parents can form or join groups based on specific topics or preferences.
- Parents can engage in a forum where they can ask questions.
- Parents can watch live streams and join live discussions.

1.4.3 Experts' section

- Experts can live stream to discuss a specific topic.
- Experts can post articles.
- Experts can provide answers to questions in the forum.
- Experts can offer downloadable resources such as e-books, worksheets, or toolkits.
- Experts can post announcements about their events or workshops.

- Etc...

1.4.4 Administration section

- Admins can monitor users' accounts.
- Admin can approve experts' accounts.
- Admins can approve articles written by experts for publication on the app.
- Admins can respond to user feedback.

1.5 Non-functional Requirements

- Availability: The app is available all the time.
- Privacy: Parents have the choice to post anonymously without sharing their personal data.
- Fault Tolerance: The app should have a backup system in place to minimize downtime.
- Compatibility: The app can function across a range of different devices.
- UI/UX design: Intuitive and user-friendly UI to enhance the overall user experience.
- Performance: The app should load quickly and respond promptly to user interactions.
- Documentation: The app includes documentation that facilitates understanding and using it.
- Multilingual support: The app can support multiple languages.

1.6 Working Methodology

1.6.1 Scrum Team

In our Scrum team, we operate with a rotating structure where each sprint, one member takes on the role of the product owner, responsible for representing the stakeholders' interests and prioritizing the backlog. Meanwhile, another team member serves as the Scrum Master, facilitating smooth sprint execution and ensuring adherence to Scrum principles and practices. Additionally, all team members actively contribute to the design of the database, leveraging their collective expertise to create efficient and scalable data structures. Similarly, everyone participates in both back-end and

front-end implementation tasks, fostering a collaborative environment where each member brings their unique skills to deliver a cohesive and high-quality product.

1.6.2 Tools Used

To adhere closely to the Scrum method's steps, we employed a project management tool specifically designed for this purpose. This facilitated effective communication among the Scrum Master, Product Owner, and development team, allowing everyone to access it at any time to monitor project progress and address any blocking factors.

Jira Software

Jira Software by Atlassian was an ideal choice for our project because it offers robust features tailored to Agile methodologies like Scrum. Its flexibility allowed us to customize workflows and boards to align perfectly with our team's needs. With Jira, we could seamlessly manage our backlog, sprint planning, and task tracking, ensuring transparency and alignment throughout the development process.

Additionally, Jira's collaborative features facilitated effective communication among team members, the Scrum Master, and the Product Owner, enhancing productivity and coordination. Its extensive reporting capabilities provided valuable insights into our project's progress, helping us identify potential bottlenecks and make data-driven decisions.

Overall, Jira Software streamlined our project management processes and contributed to the success of our Scrum team.

Visual Online Paradigm

We decided on Visual Online Paradigm as software for modeling diagrams for our project due to its intuitive communication, real-time collaboration capabilities, flexibility for updates, and inclusivity for all team members. This approach allows for clearer comprehension of complex information, enables seamless teamwork despite geographical barriers, ensures easy adaptability to evolving project needs, and promotes active participation from all stakeholders, contributing to a more cohesive and effective project environment.

Github

We chose GitHub for our project because it offers a comprehensive platform for version control, collaboration, and project management. Its robust features enable us to track changes, manage tasks, and coordinate with team members efficiently. Additionally, GitHub's integration with various development tools streamlines our workflow and enhances productivity. Furthermore, its accessibility and widespread use in the software development community ensure seamless collaboration and knowledge sharing. Overall, GitHub provides the necessary infrastructure to support our project's development process effectively.

Chapter 2

Development

2.1 First Sprint

After defining the context, definition, and objectives of our project, we then set a scope for it and gathered the functional and non-functional requirements for our application thanks to the contributions of our product owner (creation of the product backlog). Upon further discussion with the latter, we chose an order of priority for said requirements (also known as sprint planning). The conclusion for this step is that the features with the highest level of importance are those revolving around creating and logging into user accounts. To develop them, they were organized into tasks, each one assigned to a different team member. This is what is known as a sprint backlog in SCRUM. It outlines the specific tasks and activities in a sprint for a project team.

Thanks to Jira's clear and organized interface, we were able to plan the first sprint of our development project as follows:

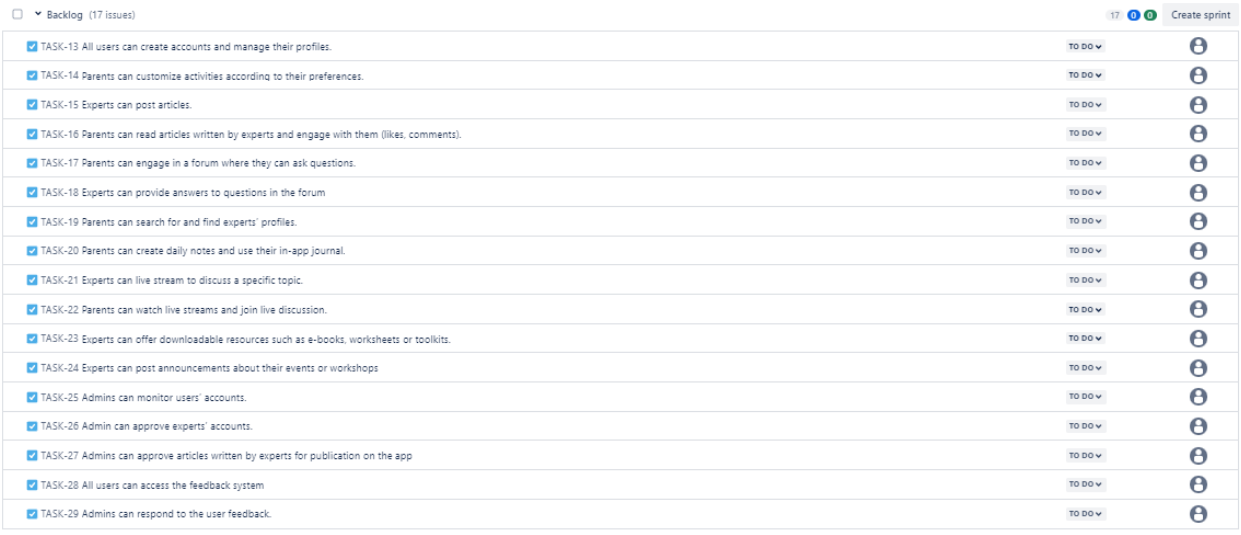


Figure 2.1: Product Backlog

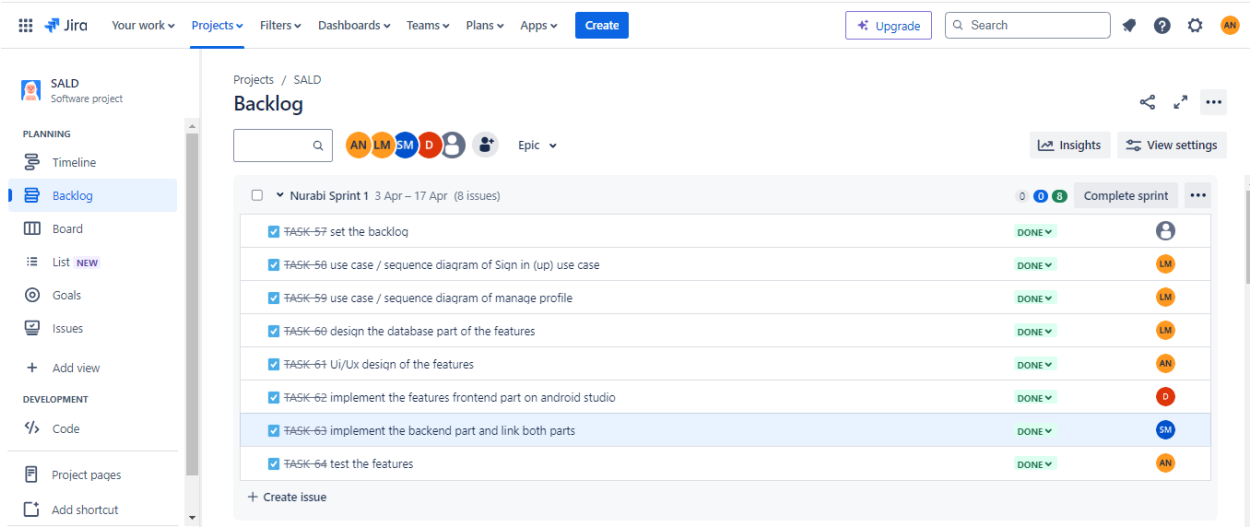


Figure 2.2: Sprint 1 Backlog

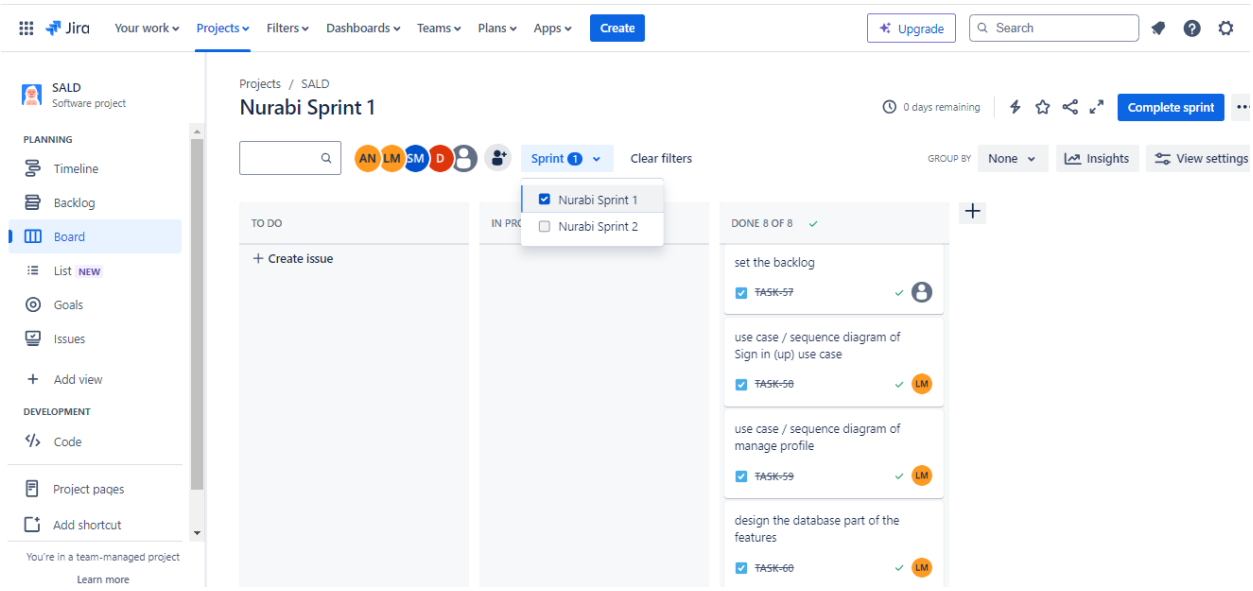


Figure 2.3: Sprint 1 Board

This sprint lasted two weeks (from April 3rd to April 17th) with daily meetings taking place to ensure communication, coordination, transparency, adaptation, and team cohesion for effective project management. At the end of the two-week period, the team gathered around for a sprint review, which is where the team showcases the work completed during the sprint to stakeholders, collects feedback, and collaborates on adjusting the product backlog as necessary for the next sprint.

Below are the sprint deliverables (documents and code that result from a sprint) that resulted from this first sprint:

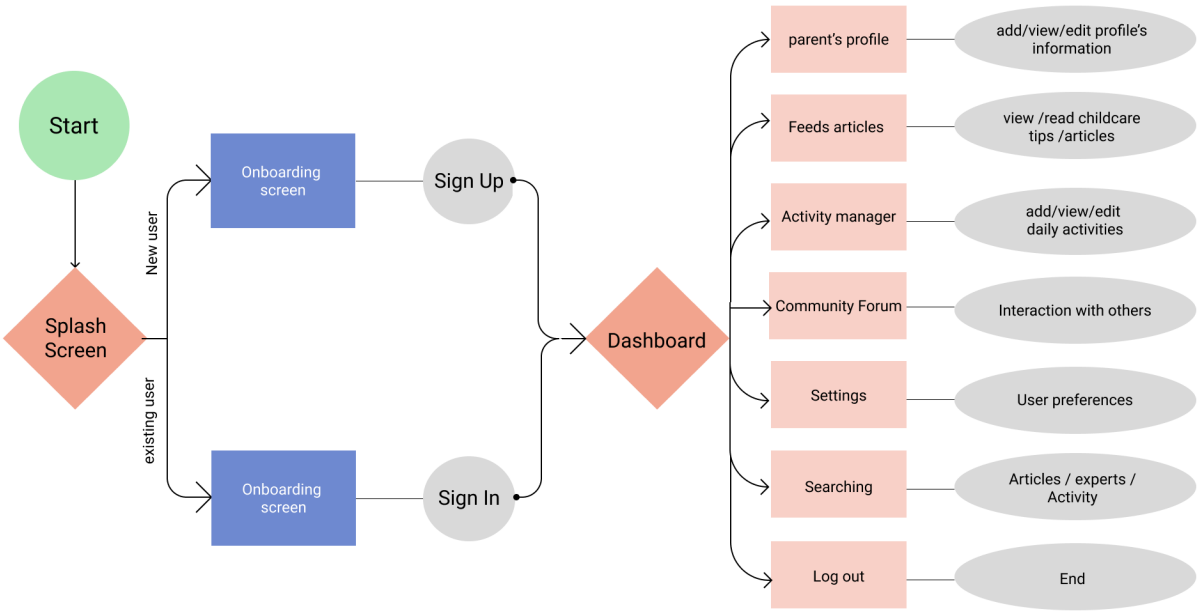


Figure 2.4: User Persona Map



NuRaBi

You are not just raising a child
you are shaping a future

Get Started

Log in

Figure 2.5: Home Screen UI Prototype

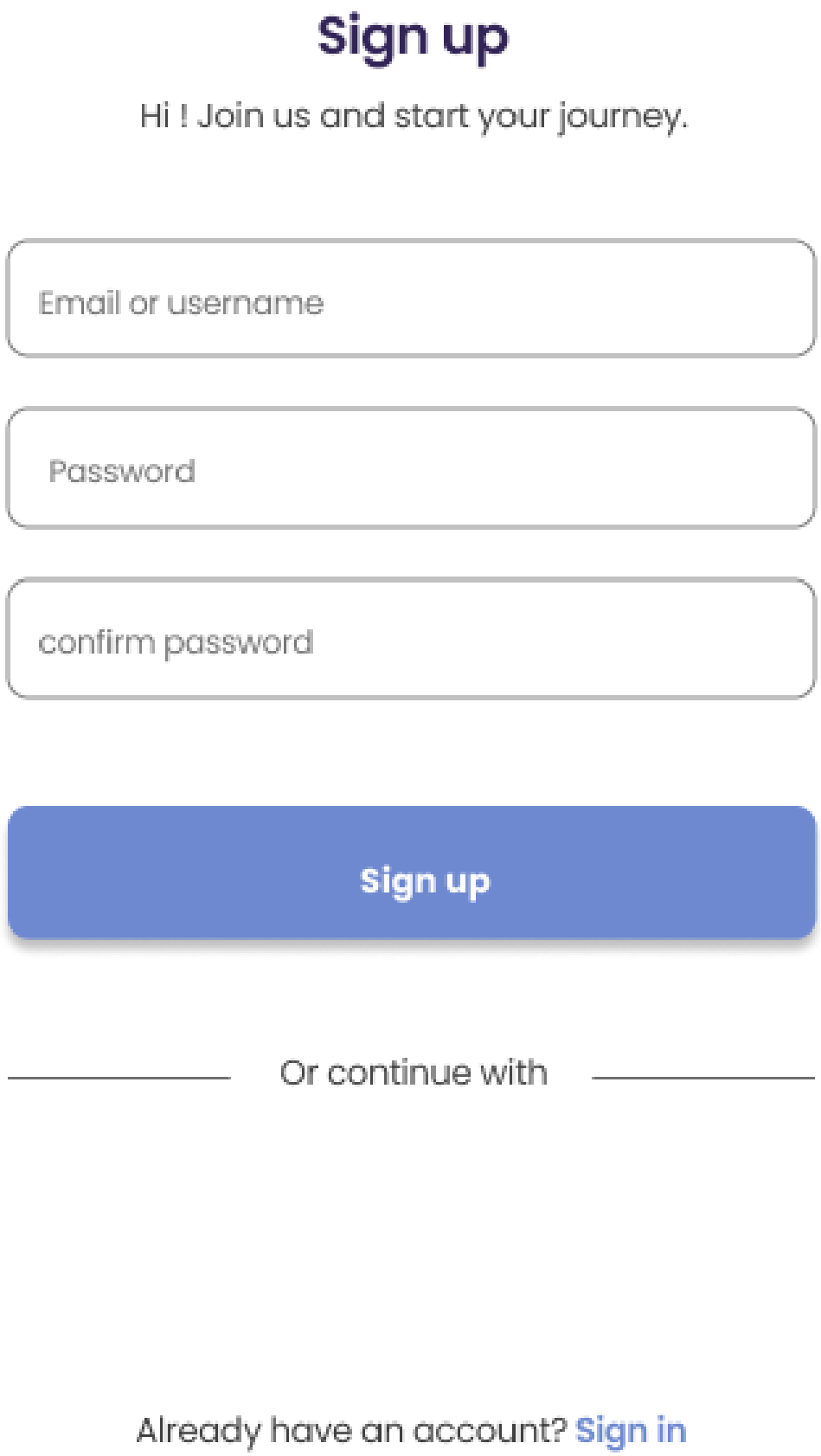


Figure 2.6: Sign Up UI Prototype

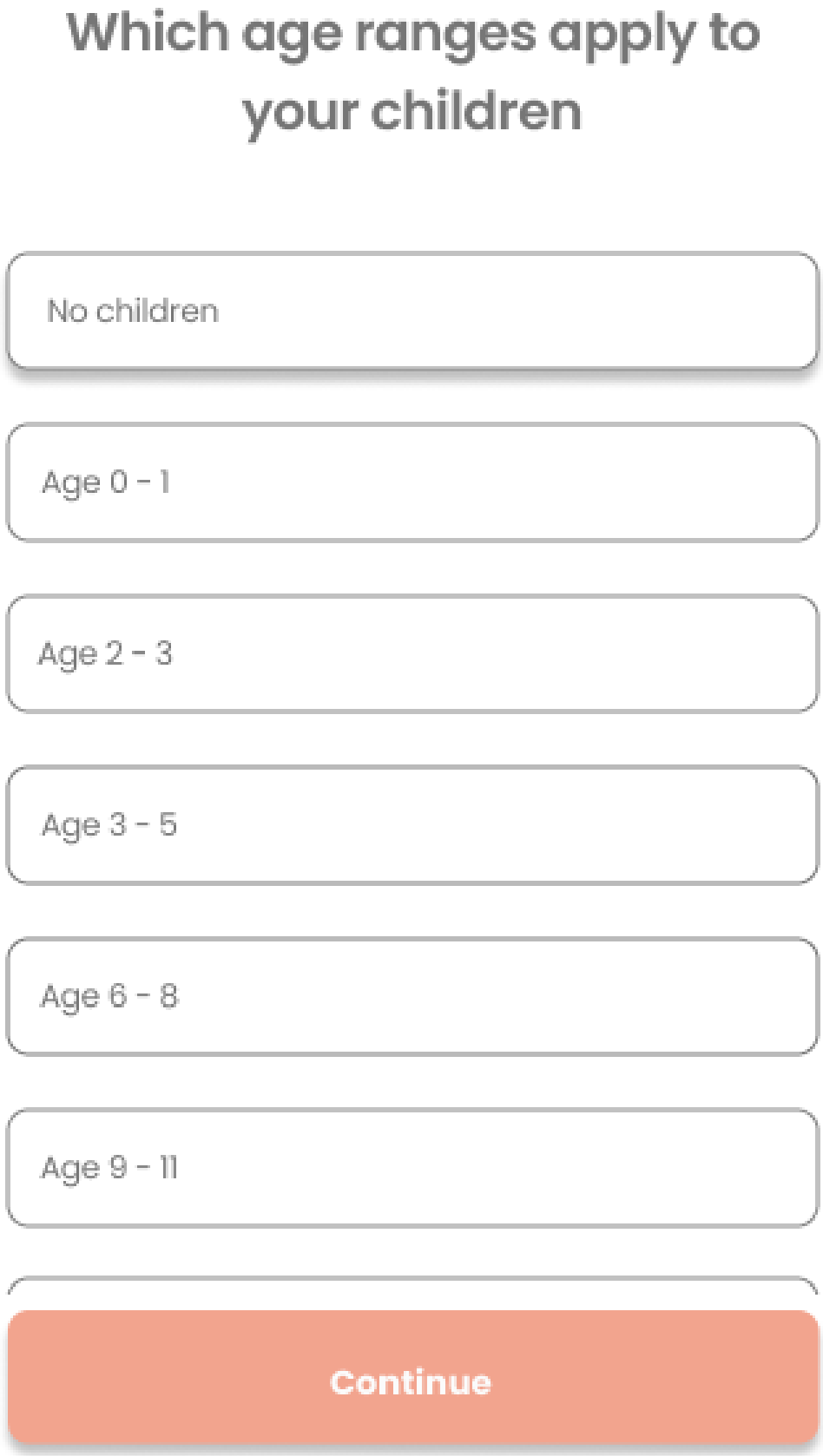


Figure 2.7: Sign Up Questionnaire

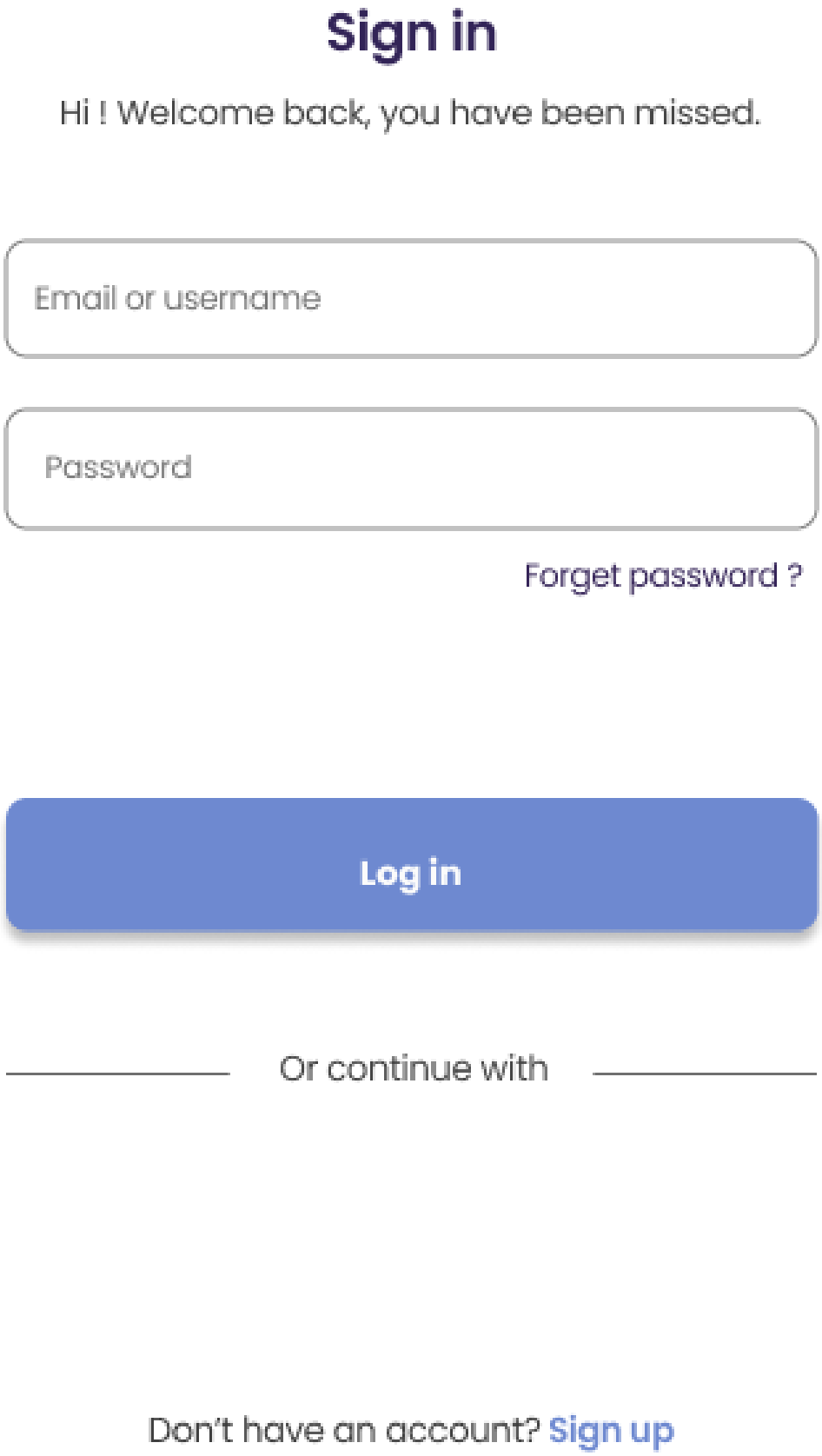


Figure 2.8: Sign-In UI Prototype

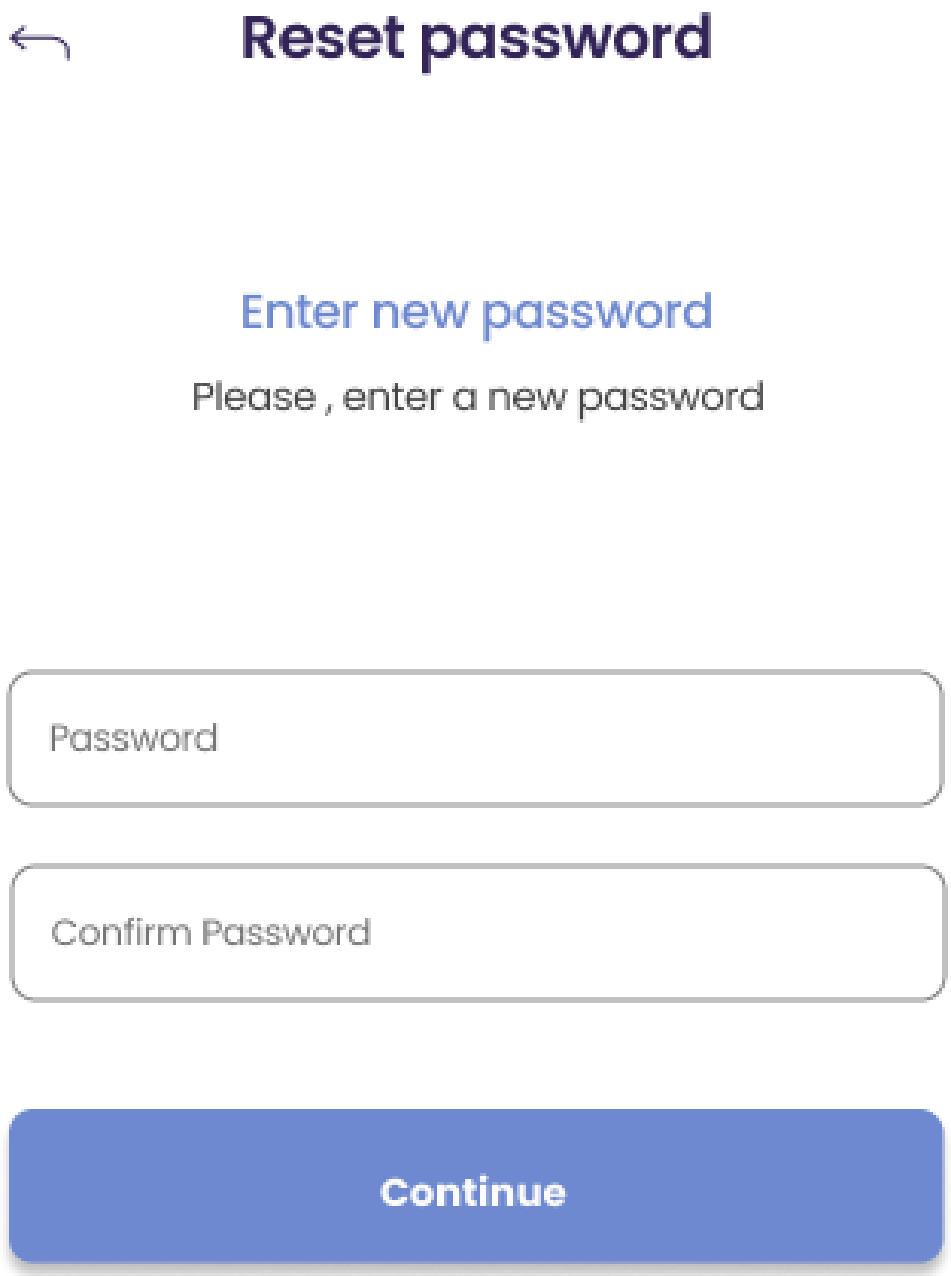


Figure 2.9: Reset Password UI Prototype

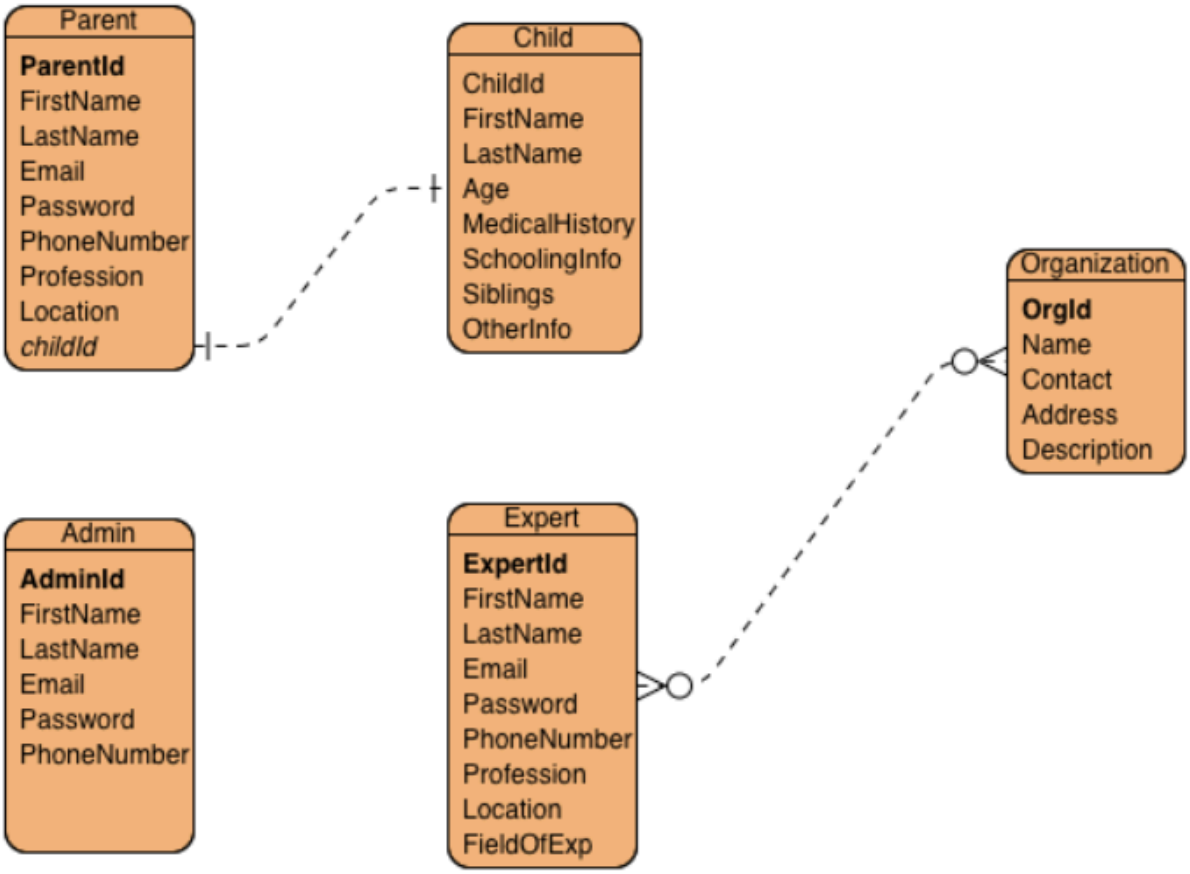


Figure 2.10: Entity-Relationship Diagram

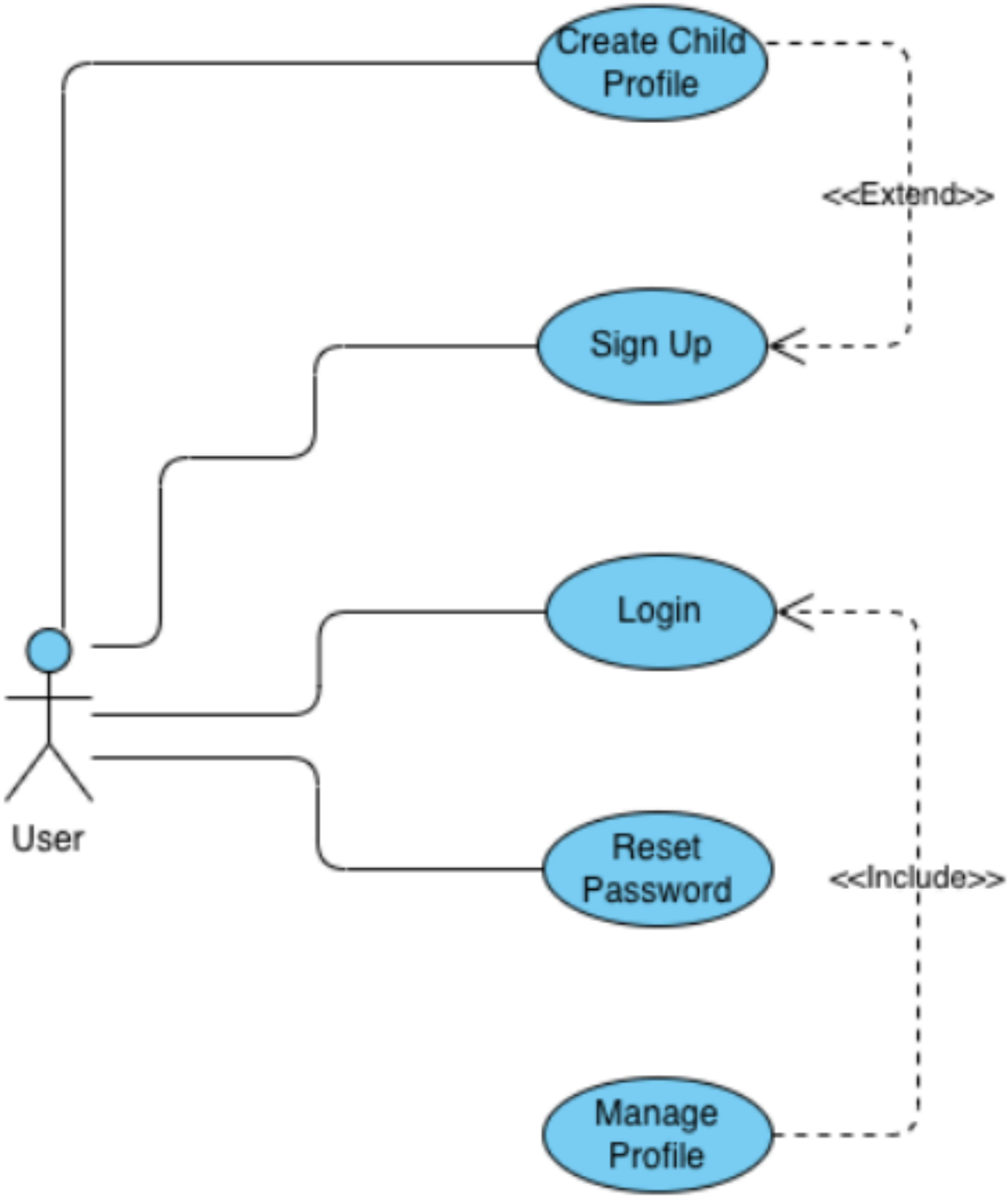


Figure 2.11: Use Case Diagram

Use Case	Login
Type	Primary
Participating Actors	Parent, Expert, Admin
Objectives	Allows registered users to log into their accounts (authentication).
Pre-conditions	The user must have an account.
Basic Flow	1. The user presses the “Login” button. 2. The app displays a login form (email and password fields). 3. The user fills in his information and presses the “Login” button. 4. The system verifies his login credentials and logs him in.
Alternative Flow	Starts after step 3 of the basic flow: 1. The system doesn’t find the login information in the database and displays an error message.
Exception Flow	/
Termination Outcome	The user is logged in.

Figure 2.12: Login Specifications

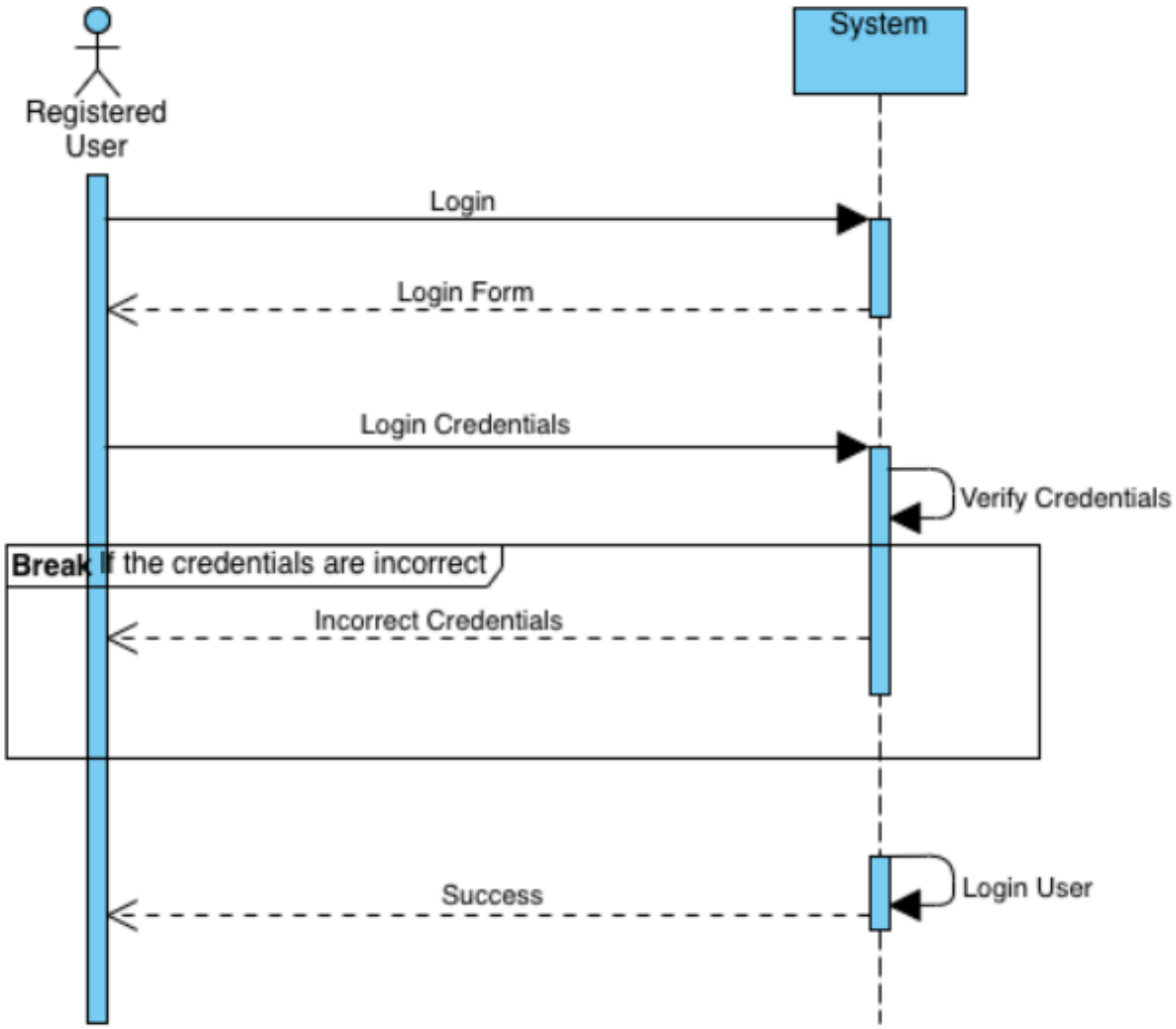


Figure 2.13: Login Sequence Diagram

Use Case	Sign-Up
Type	Primary
Participating Actors	Unregistered User
Objectives	Allows unregistered users to create an account.
Pre-conditions	/
Basic Flow	1. The unregistered user presses the "Sign-Up" button. 2. The system displays a sign up form (first and last name, email, password, password confirmation, user name, parent/guardian status…). 3. The unregistered user fills in his information and presses the "Sign-Up" button. 4. The system verifies his sign up information and creates an account for him.
Alternative Flow	Starts after step 3 of the basic flow: 1. The system finds an error in the entered information and displays an error message.
Exception Flow	/
Termination Outcome	The account is created.

Figure 2.14: Sign-up Specifications

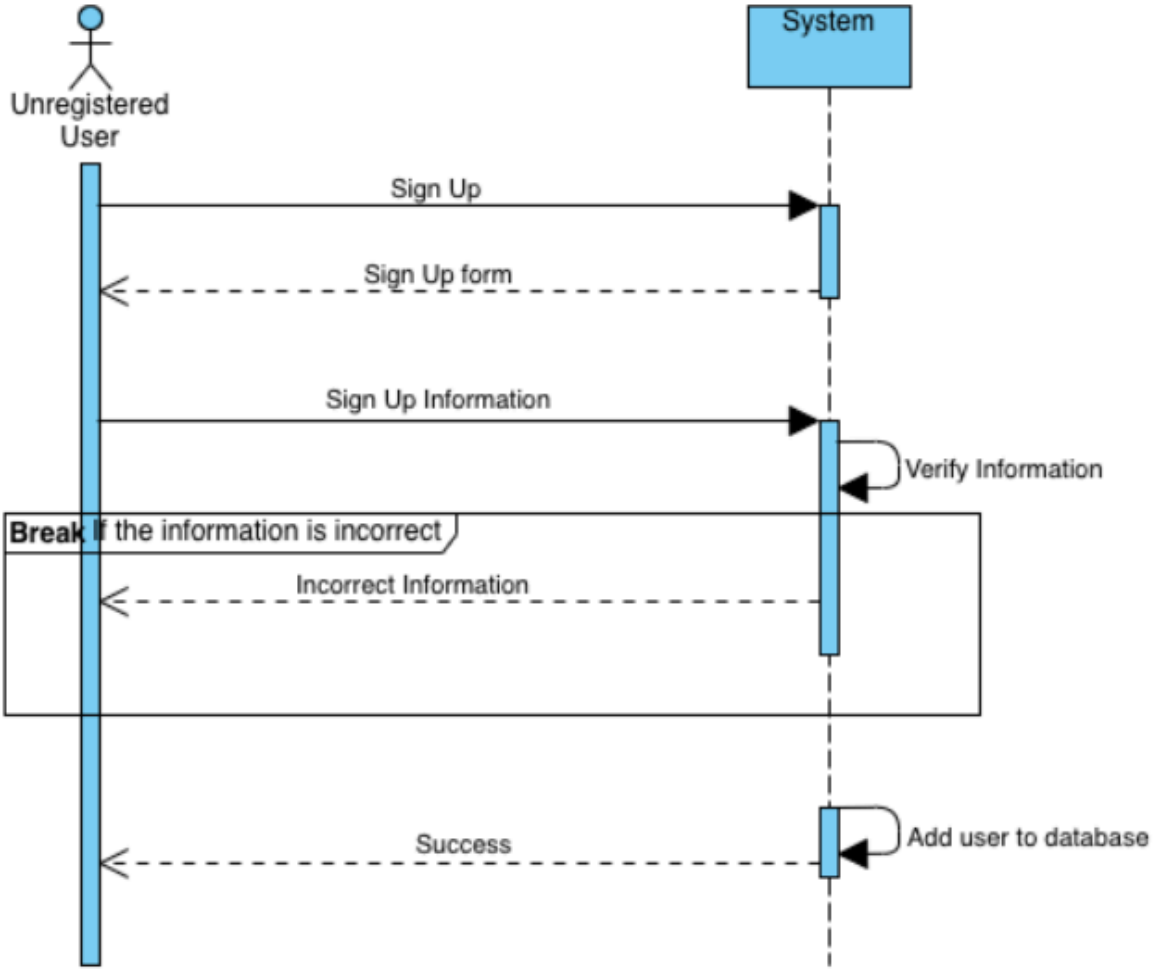


Figure 2.15: Sign Up Sequence Diagram

Use Case	Create Child Profile
Type	Secondary
Participating Actors	Parent
Objectives	Allow parents to create a profile for their child, providing relevant information.
Pre-conditions	• The parent must be logged into their parent account. • The parent must not have another child profile.
Basic Flow	1. Parent navigates to the "Child Profile" or "Add Child" section within the app. 2. System presents a form for the user to input their child's information. 3. Parent enters the following details about the child: • Name • Gender • Date of Birth • Health History (if applicable) • Educational Performance (if applicable) • Number of Siblings • Other relevant details 4. Parent submits the completed form. 5. System validates the entered information. 6. System creates a new child profile with the provided information. 7. System confirms successful creation of the child profile.
Alternative Flow	/
Exception Flow	Starts after step 5 of the basic flow: 1. If there are any errors or missing required fields, the system displays error messages and prompts the parent to correct them.
Termination Outcome	Parents successfully create a profile for their child, providing relevant information for personalized parenting assistance and guidance.

Figure 2.16: Create Child Profile Specifications

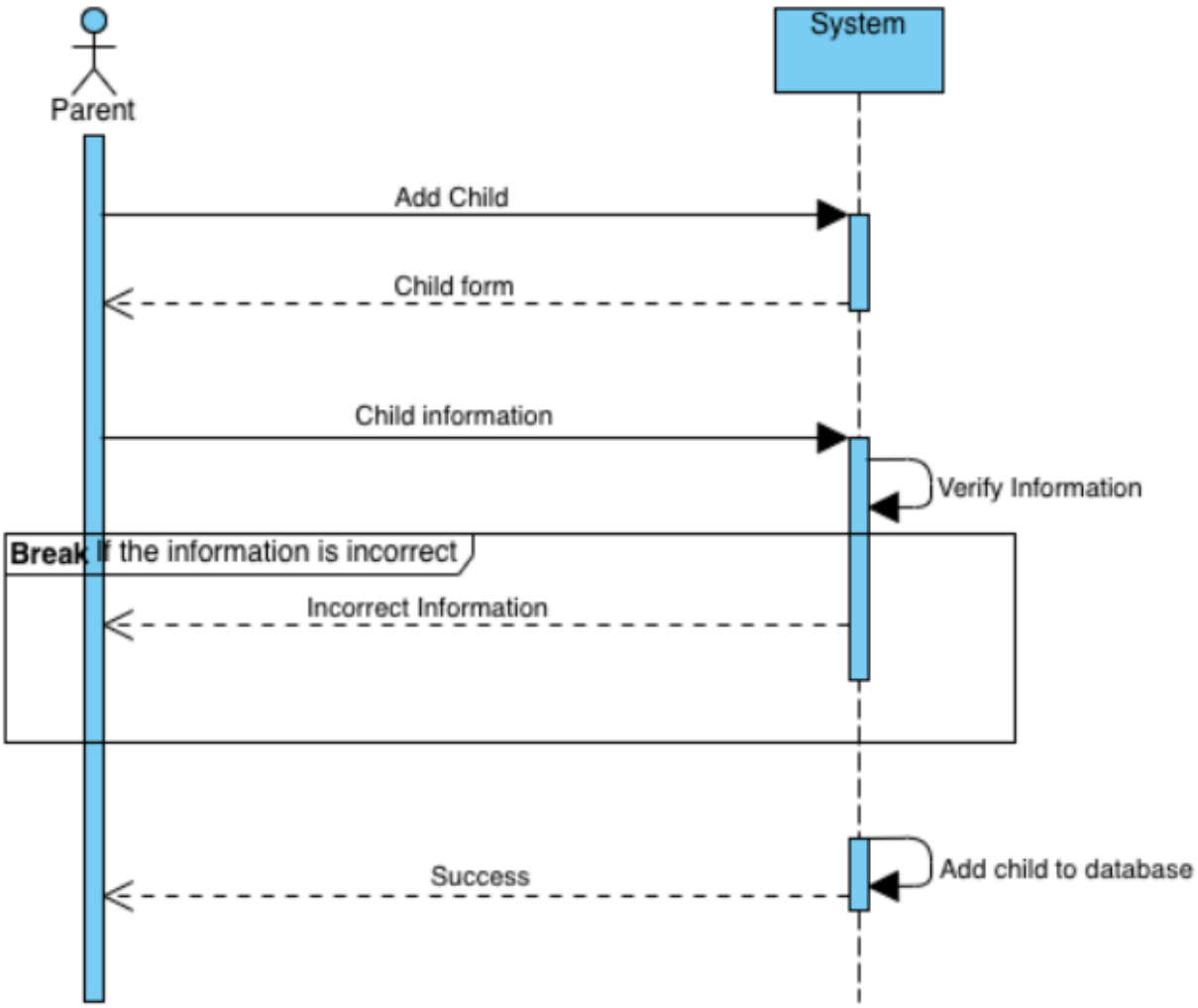


Figure 2.17: Create Child Profile Diagram

Use Case	Reset Password
Type	Secondary
Participating Actors	Parent, Expert, Admin
Objectives	Allow users to reset their password if they have forgotten it or wish to change it for security reasons.
Pre-conditions	• The user must have previously registered an account. • The user must have access to the email address associated with their account.
Basic Flow	1. User selects the "Forgot Password" option on the login screen. 2. System prompts the user to enter their email address. 3. User enters their email address and submits the form. 4. System validates the email address. 5. System generates a unique password reset link and sends it to the user's email address. 6. User checks their email and clicks on the password reset link. 7. System verifies the validity of the password reset link. 8. User enters a new password and confirms it. 9. System updates the user's password with the new one and confirms successful password reset.
Alternative Flow	Starts after step 4 of the basic flow: 1. If the email address is not valid or not associated with an account, display an error message and allow the user to retry or contact support.
Exception Flow	Starts after step 7 of the basic flow: 1. If the link is valid and has not expired, the system prompts the user to enter a new password.
Termination Outcome	User successfully resets their password and gains access to their account.

Figure 2.18: Reset Password Specifications

Use Case	Manage Profile
Type	Primary
Participating Actors	Parent, Expert, Admin
Objectives	Allow users to edit their personal information.
Pre-conditions	• The user must be logged into their account. • The user must have permission to edit their profile.
Basic Flow	1. User navigates to the "Profile" or "Account Settings" section within the app. 2. System displays the user's current profile information, including name, email address, and any other relevant details. 3. User selects the option to edit their profile. 4. System presents a form with editable fields for the user's profile information. 5. User makes changes to their profile information as desired. 6. User submits the updated profile information. 7. System validates the updated information. 8. System updates the user's profile with the new information. 9. System confirms successful profile update.
Alternative Flow	/
Exception Flow	Starts after step 7 of the basic flow: 1. If there are any errors or missing required fields, the system displays error messages and prompts the user to correct them.
Termination Outcome	User successfully updates their profile information with the desired changes.

Figure 2.19: Manage Profile Specifications

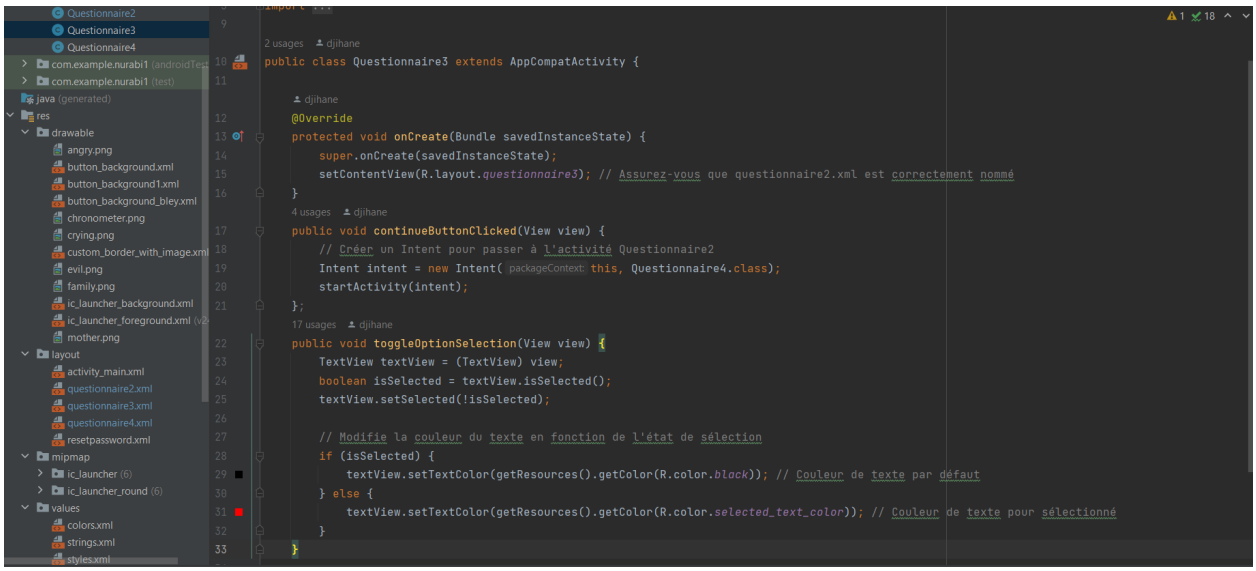


Figure 2.20: Code Snippet 1

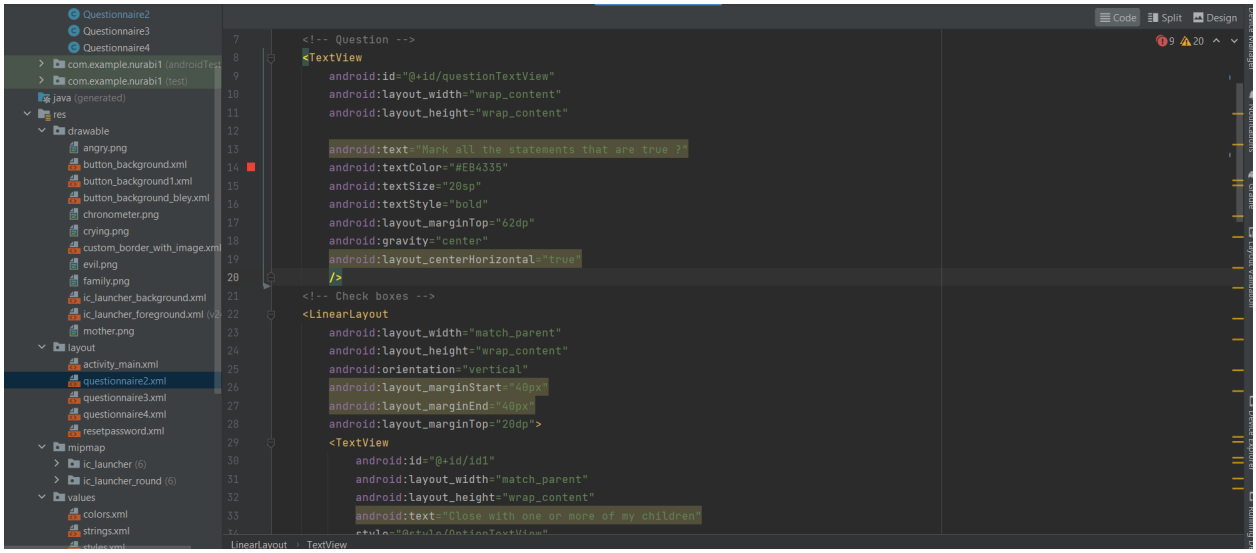


Figure 2.21: Code Snippet 2

As mentioned earlier, a sprint review was held to evaluate the team's work and progress. We concluded that some adjustments needed to be made in order to meet the product owner's expectations (talk about adjustments). We also took this opportunity to plan ahead for the next sprint which would be focused on developing the parents' main features, namely the daily exercises and journal tabs.

2.2 Second Sprint

Immediately after the first one, another two-week-long sprint was initiated from April 18th to May 3rd. This time, the team members switched roles as explained in the Scrum Team section. With a fresh outlook, new user stories were written and the subsequent tasks were assigned to our members (refer to the screenshots below).

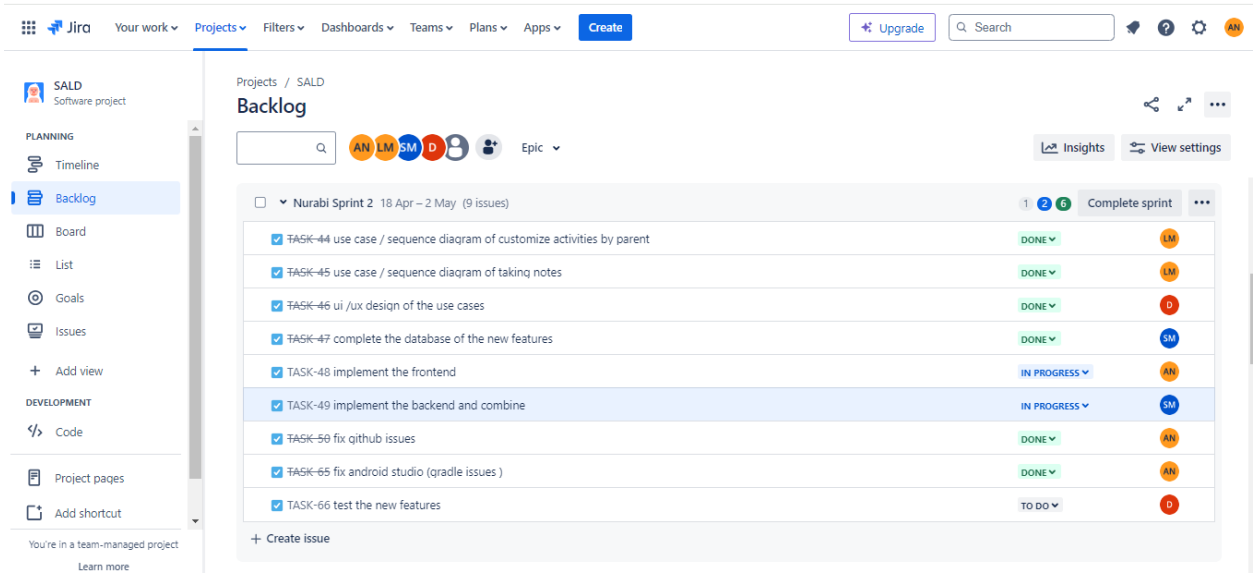


Figure 2.22: Sprint 2 Backlog

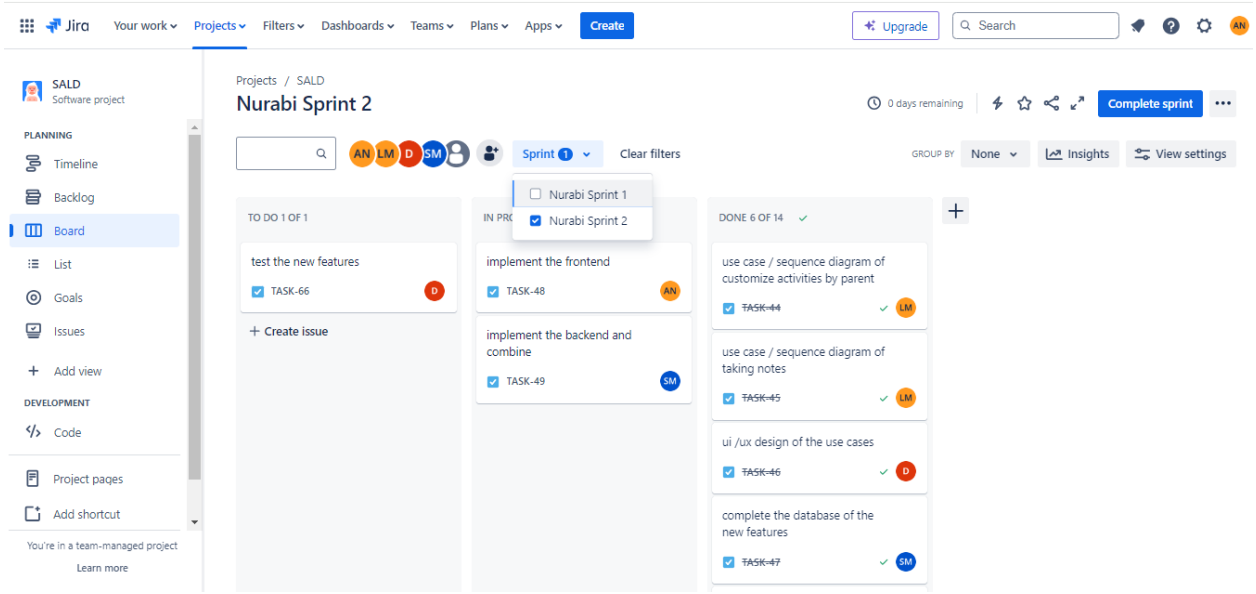


Figure 2.23: Sprint 2 Board

Once again, the team members gathered at the end of every day of the week to discuss their progress. At the end, they got together one final time for the sprint review to go over everything

they worked on. Below are this sprint's deliverables:



Figure 2.24: Parent Dashboard UI Prototype

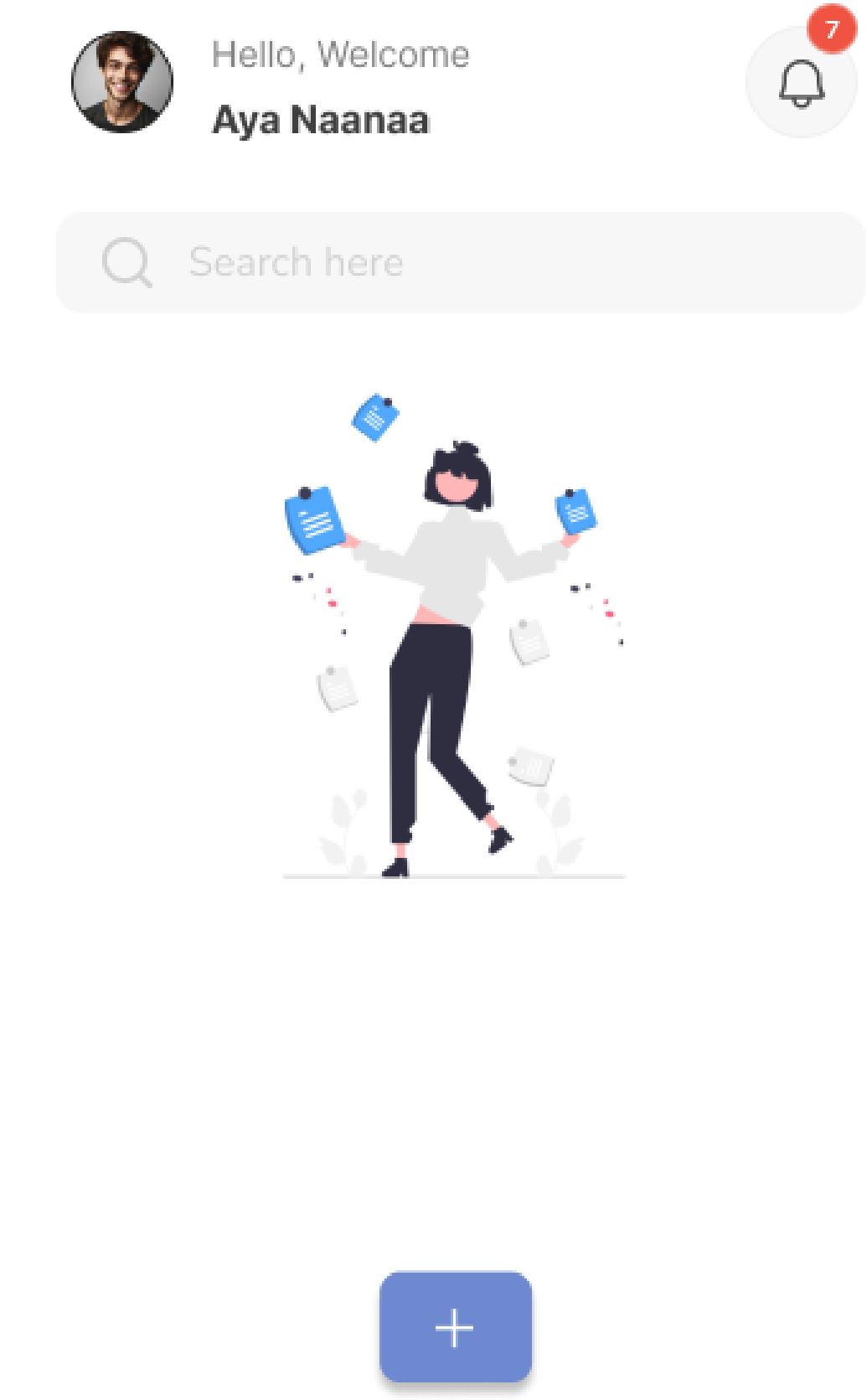


Figure 2.25: Journaling UI Prototype 1

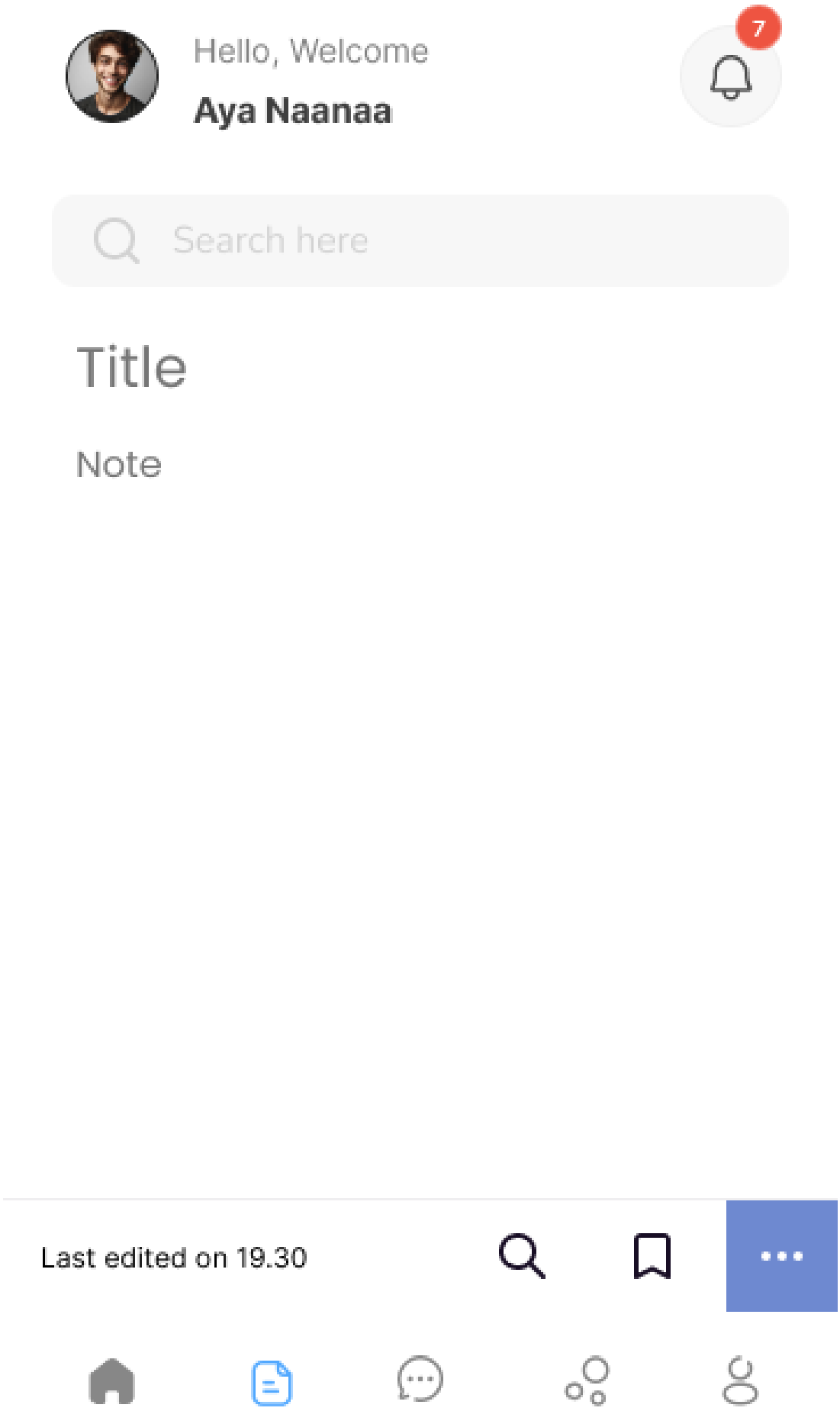


Figure 2.26: Journaling UI Prototype 2

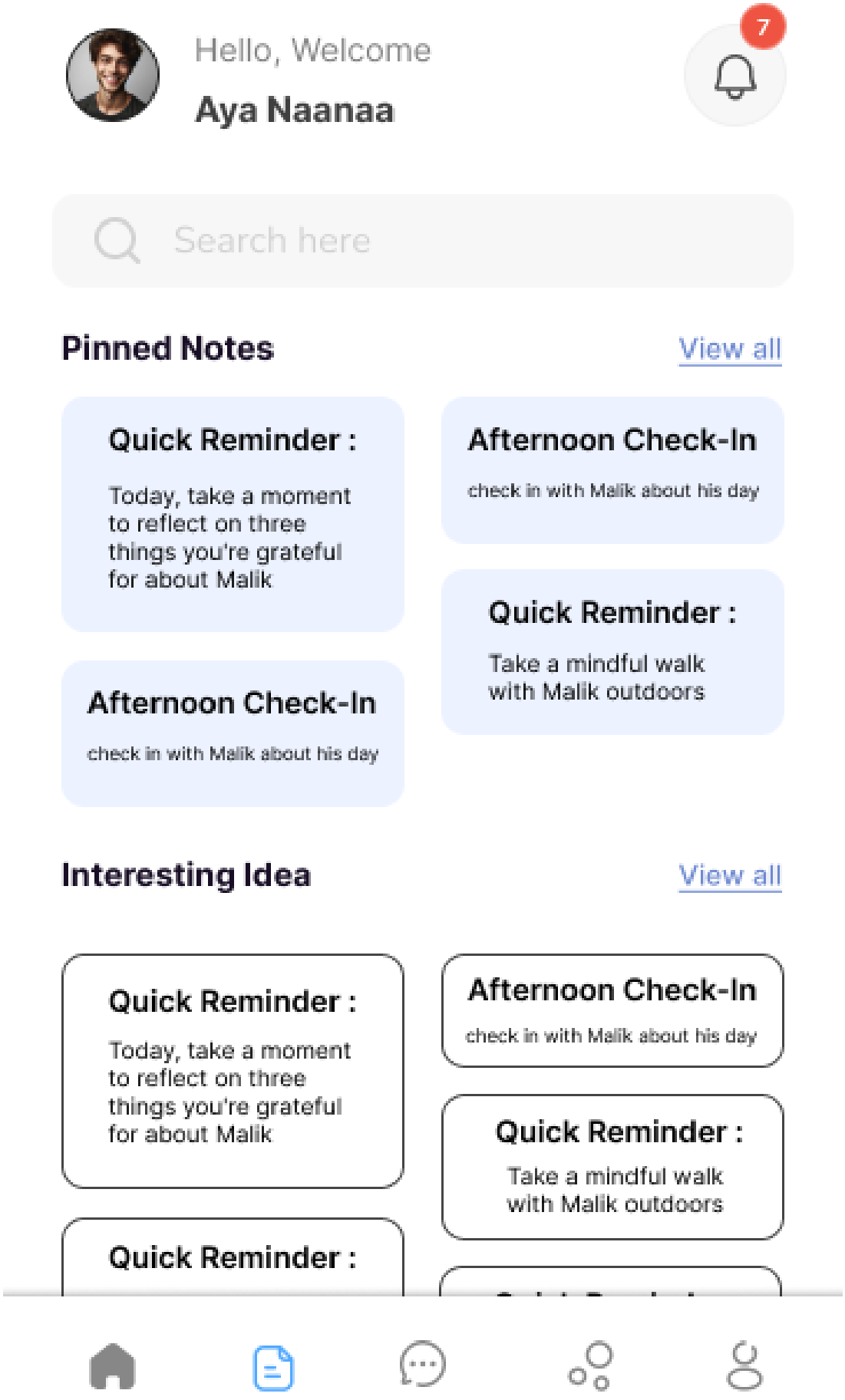


Figure 2.27: Journaling UI Prototype 3

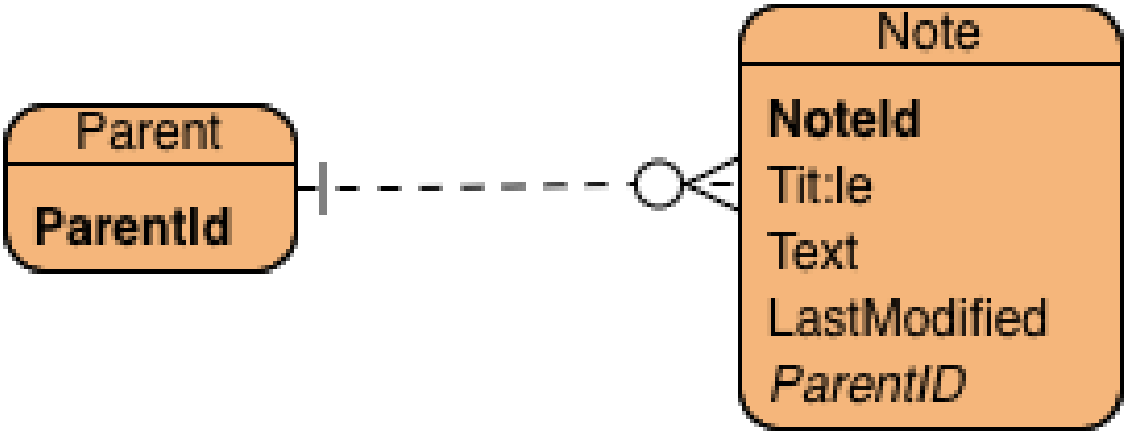


Figure 2.28: Entity Relationship Addition

Use Case	Add to Journal
Type	Secondary
Participating Actors	Parent
Objectives	Allow parents to create notes in their parenting journal.
Pre-conditions	• The parent must be logged into their parent account.
Basic Flow	1. The parent navigates to the journal tab from the parent dashboard. 2. The parent presses the add new note button. 3. A screen with a title field and body text field is shown. 4. The parent fills those fields as necessary. 5. The system saves the note automatically when the parent navigates to another section.
Alternative Flow	/
Exception Flow	/
Termination Outcome	Parents successfully add a new note to their journal. The system saves it in the database.

Figure 2.29: Add to Journal Specifications

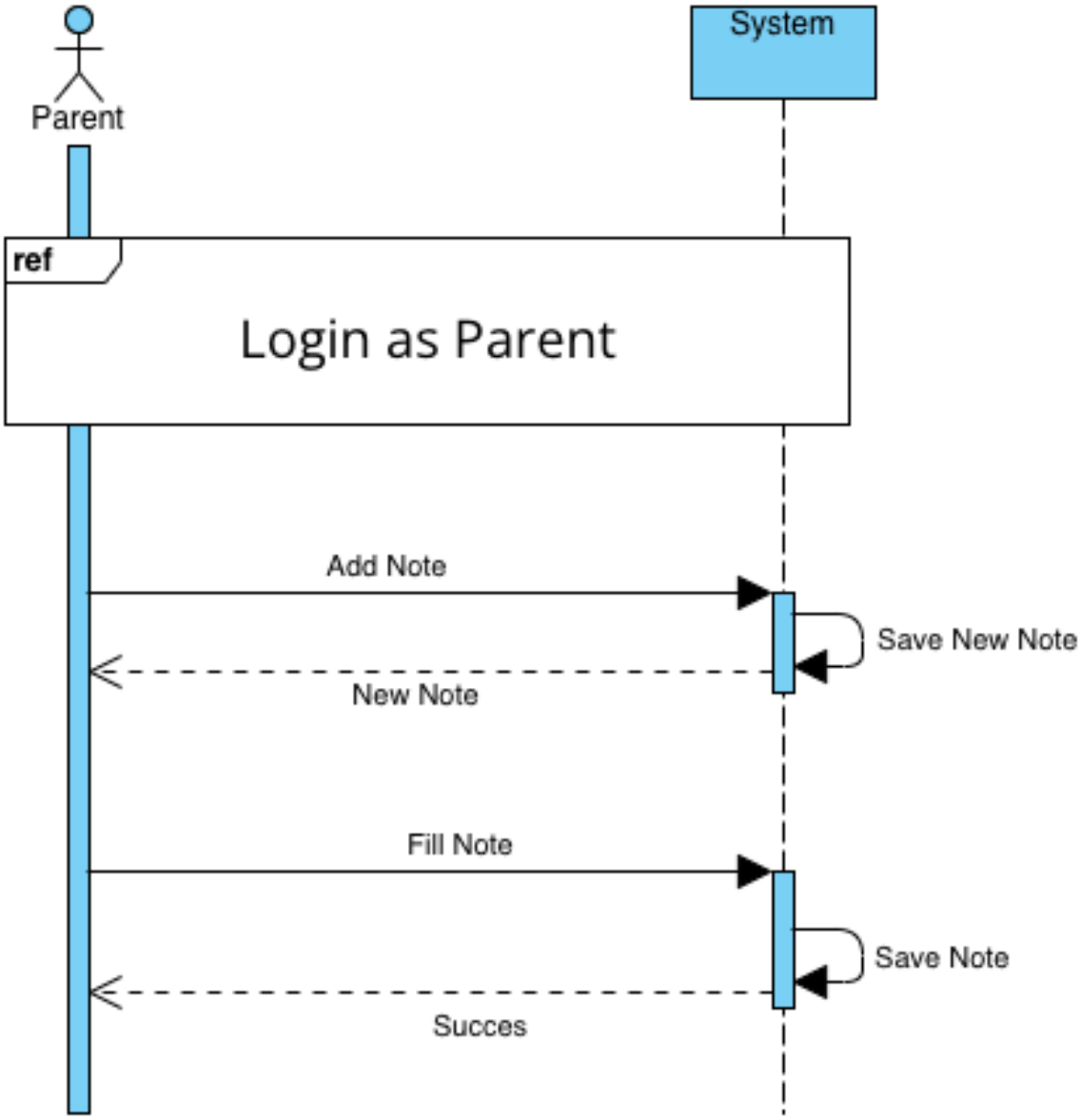


Figure 2.30: Add to Journal Sequence Diagram

Conclusion

After the completion of both sprints, the product owner deemed it appropriate to bring the project to a temporary close thus eradicating the need for a 3rd sprint for now.

All members of the team were satisfied with their work, as SCRUM allowed them to keep a good momentum and gave them clear and concise tasks that they could achieve in minimal time. This approach not only accelerated development but also made it more efficient.

In the future, our team will simply pick up where they left off by iterating through more sprints to continue developing the app. Herein lies the true power and beauty of SCRUM.